

CHAPTER 1

INTRODUCTION

A manipulator is, in its simplest description, a chain of rigid links connected by powered joints, ending in a tool that does the actual work of picking something up, placing it, or holding it against a process. For most of industrial robotics' history this chain worked alone, fenced off from people, repeating a single memorised path thousands of times a shift. The collaborative arm changes that arrangement: it is built to share a workspace with a person, which means every motion it makes must be treated as something that could come near a hand, not just near a workpiece. This thesis is built around one specific arm of that kind, and the platform is not incidental to the argument. Everything the simulation trains against, from the joint speeds it is allowed to command to the distances it is allowed to close, is set to match what this particular machine can actually do.



Figure 1.1. A UR5e collaborative arm sharing a workspace with a person. Photograph reproduced from Xia et al. [?], Fig. 1, under that paper's CC BY-NC-ND 4.0 licence. Placeholder pending an original photograph of the platform used in this project.

The platform is a Universal Robots UR5e, a six-joint (6-DOF) collaborative arm [?]. It carries a payload of 5 kg to a reach of 850 mm, repeats a taught position to within 0.03 mm,

and weighs 20.6 kg unmounted. Every one of its six joints is rated to the same maximum speed, 180 deg/s, which is exactly π rad/s. The simulated arm in this thesis is capped at 3.14 rad/s for the same reason a real one is: past that speed the motion stops being something a person nearby could react to in time. The provenance of that figure matters: 3.14 rad/s is the datasheet limit itself, carried into the simulator so that a policy trained there is never asked to move faster than the physical machine could follow it. The UR5e also ships with seventeen configurable safety functions and is certified to the relevant collaborative-robot safety standards, ISO 13849-1 (Category 3, Performance Level d) and ISO 10218-1 [?]. None of those seventeen functions is exercised by the simulation. What this thesis borrows from the platform is its physical envelope, its speed and reach limits, not its built-in safety controller, and Chapter 3 is explicit about which of the two is under test.

Machines with this specification already do real work outside the laboratory. UR-class arms in the UR5e's size class are common on production floors for exactly the tasks a 5 kg payload and an 850 mm reach suit: pick-and-place, machine tending, small-parts assembly and packaging, and inspection stations where a part has to be picked up and held in front of a camera. The same collaborative certification that lets one work next to a person on a line is also why it turns up so often in research: it is affordable enough for a university laboratory, safety-rated enough to run near people without a fenced cell, and common enough on real production floors that a result obtained on one arm should say something useful about the rest of that population. Two of the papers already cited in this chapter used a UR5-class arm for exactly that reason, one for grasping [?] and one for collision avoidance around a moving person [?]. The platform used here is drawn from that same working population.

Two further reasons fixed the choice. The preceding project in this laboratory used a different, custom five-degree-of-freedom arm for classical visual servoing [?]; this thesis moves to a UR-class platform because the safety question it asks, how a learned policy behaves near a joint limit or a kinematic singularity, is best posed on hardware whose limits and safety record are already public and well documented. Isaac Lab, the simulation framework used throughout (Chapter 3), also ships the UR5e as one of its reference manipulators, so building the environment on this arm meant working from a supported example rather than adapting a custom robot from scratch. The joint limits and speed ceiling used throughout this thesis therefore belong to a machine that exists, not one assumed for convenience.

1.1 Background

Industrial manipulators increasingly work in spaces shared with people rather than behind a fence, and the UR5e described above is a working example of that shift. A controller that follows a fixed, hand-derived trajectory can be inspected and bounded before it ever moves: an engineer can trace every configuration it will visit and check each one against

a joint limit or a known obstacle. A policy obtained by training a reinforcement learning agent cannot be inspected the same way. Its behaviour is known only through the statistics of the trajectories it produces during and after training, not through a proof that covers every configuration it might reach. Safe operation of a UR-class collaborative arm around people is already treated as a live industrial requirement, not a hypothetical one [?]. Reinforcement learning is attractive for manipulation precisely because it can learn grasping and placement behaviour directly from experience, without a hand-built controller for every object and pose the arm might encounter. What that learned behaviour does in the configurations it visits rarely, however, is not a detail to be checked after the fact. For a policy sharing a workspace with a person, it is the condition deployment depends on.

The standard formulation of reinforcement learning gives a designer only one channel for stating what the agent should avoid: the reward function. Take the UR5e’s own joint-speed ceiling as a concrete case. A designer who wants the trained policy to respect it has, in the standard formulation, exactly one lever: add a penalty term for exceeding it, sized by a weight that trades a unit of excess speed against a unit of task reward. That weight has to be fixed before training starts, before the designer has any real sense of how large a violation the untrained policy will produce or how much task performance the penalty will cost once it starts to bind. Set the weight too small and the constraint is decorative. Set it too large and the agent may refuse to move at all. An alternative, pursued in this thesis, is to keep the requirement outside the reward entirely and state it instead as a constraint with an explicit budget: a quantity that can be set from a measurement of the system, such as the arm’s own rated limits, and checked afterwards against what the trained policy actually used [?]. The distinction matters because of what each side has to answer for. A reward weight is a guess, tuned by trial and error against whatever the agent happens to do. A constraint budget is a number the designer can defend before training ever starts, because it comes from a measurement rather than a hunch. Chapter 2 reviews this distinction, and the literature built on either side of it, in full.

1.2 Problem Description

This thesis puts that alternative to a direct test. It trains a UR5e in simulation to pick up and lift a cube, comparing a standard proximal policy optimisation (PPO) agent against a constrained variant of the same algorithm (PPO-Lagrangian) under a shared floor on manipulability, a measure of how far a configuration sits from a kinematic singularity. Such a comparison, as usually run in the literature, has two problems.

- **The cost-critic confound.** Adding a constraint to an RL agent does not add only a constraint. A Lagrangian method also adds a cost critic, a second value network trained alongside the policy to estimate the constraint violation the current policy would incur.

That network is present whether or not the constraint is actively binding, and its presence can influence training through machinery the two agents no longer share. A comparison that trains only the unconstrained baseline and the fully constrained agent cannot separate what the constraint achieved from what the extra network alone changed.

- **Seed variance.** A single training run under one random seed says little about what a method will do on the next run [?]. Published comparisons of constrained against unconstrained manipulation policies typically report one run, or a mean over few, without a measure of how much the outcome would vary if the same method were retrained.

Section 2.11 sets out this gap in full against the closest published comparison [?]. What matters here is the consequence for design. Chapter 3 describes a comparison built to answer both problems at once. Alongside the unconstrained baseline it places a control arm, identical to the constrained agent in every respect except that its Lagrange multiplier is pinned to zero, so that whatever the cost critic does on its own can be measured apart from whatever the constraint does. The constrained agent is trained at two episodic cost budgets, 25 and 15, because a tighter budget and a looser one make different demands of the policy. Every arm is trained on five independent random seeds, and every reported quantity carries its spread across those seeds instead of standing as a single number.

The control arm turned out to reproduce the unconstrained baseline exactly, at the level of the stored network weights rather than merely to within statistical noise, so the two are reported jointly as a single arm throughout Chapter 4. The isolation this design was built to provide therefore measures the cost critic’s independent effect as exactly zero, and Section 4.2 gives that verification in full. Three arms are reported in the end: the joint baseline, and the constrained agent at each of its two budgets.

The work also continues a line of manipulator research from the same laboratory. The preceding study [?] addressed the perception and motion side of manipulation, tracking and following a target with classical image-based visual servoing. This thesis addresses the control policy instead, with reinforcement learning, and adds a safety constraint that a classical visual servoing loop does not itself provide.

1.3 Objectives

The general objective of this thesis is to determine whether expressing a manipulability safety requirement as an explicit constraint, rather than as a term in the reward, changes the safety behaviour of a learned grasping policy on a UR5e manipulator, and whether it changes how predictable that behaviour is across independent training runs.

The specific objectives are as follows:

- To implement an unconstrained PPO baseline and a constrained PPO-Lagrangian agent for a UR5e pick-and-lift grasping task in Isaac Lab, trained under a shared floor on manipulability.
- To isolate the effect of the constraint from the effect of the cost critic that implements it, by training a control arm that carries the cost critic with its Lagrange multiplier held at zero throughout training.
- To establish whether the constrained agent’s behaviour depends on how tight its budget is, by training it at two episodic cost budgets, 25 and 15, instead of one.
- To train every arm on five independent random seeds and report task performance and safety outcomes as a distribution across those seeds instead of as a single mean.
- To evaluate every trained policy on frozen, held-out episodes, so that the reported safety and task figures describe the policy that would actually be deployed rather than a still-learning one.
- To determine, from the resulting distributions, whether the constraint changes the average safety outcome of a training run, the spread of that outcome across runs, or both.

1.4 Scope

This thesis was registered under the title *Safe Adaptive Image-Based Visual Servoing with Constrained Reinforcement Learning for Precision Grasping on a UR5 Manipulator: From Simulation to Real Hardware*, and was proposed as a three-layer programme, not as Layer 1 alone. The title on this book is narrower because the delivered work is narrower, and the paragraphs below say exactly where the boundary falls.

Layer 1, delivered here, is safe reinforcement learning for grasping in simulation, using privileged pose information rather than vision. Layer 2 would have closed the loop with vision: an image-based visual servoing scheme in which the image Jacobian relating pixel motion to joint motion is tuned by the learned policy rather than fixed in advance, replacing the privileged pose feedback of Layer 1 with an eye-in-hand RGB-D camera and object detection. It would have been a direct upgrade of the predecessor project in this laboratory, which used a fixed, hand-derived image Jacobian for the same visual-servoing task [?]. Layer 3 would have carried the resulting controller onto the physical UR5e over ROS 2, closing the gap between the Robotiq 2F-85 gripper modelled in simulation and the different gripper fitted to the laboratory’s real arm.

Only Layer 1 is delivered. Layers 2 and 3 are not attempted, and that is stated here as a deliberate narrowing of scope, decided against a fixed submission date, rather than as an

unmet target. Building an evaluation protocol able to separate a safety constraint’s effect from an implementation artefact, and to characterise it across independent seeds, could not be done to a defensible standard alongside two further systems-integration efforts of comparable size. Both dropped layers are carried forward as future work in Chapter 6.

Layer 1 on its own remains a substantial question. Constraining a policy’s behaviour with an explicit safety budget, rather than through a shaped reward, is already active practice on UR-class hardware doing collision avoidance around a moving person [?], and on other manipulator platforms doing constrained grasping specifically [?]. What this thesis contributes sits inside that same practice: a comparison protocol for measuring what the constraint itself does, isolated from an implementation artefact and reported across seeds rather than as a single run, on a platform common enough in both industry and the published literature that the result should say something beyond this one laboratory’s copy of it.

Within Layer 1, the scope is:

- **Task.** Single-object pick-and-lift grasping on a simulated UR5e in Isaac Lab, on the weld-grasp environment described in Chapter 3. The gripper is present and driven, but the moment of contact is abstracted rather than modelled as a full physical grasp.
- **Comparison.** Three algorithm arms: an unconstrained PPO baseline and two PPO-Lagrangian variants at different cost budgets, each trained on five independent random seeds.
- **Safety quantity.** A single constrained quantity, a floor on manipulability. Joint-limit proximity and workspace clearance are measured and reported but are not themselves constrained.
- **Evaluation.** The frozen policy produced by each run, assessed on held-out episodes with the multiplier and exploration noise fixed, not the training-time behaviour of a policy still being updated.
- **Algorithms.** The comparison uses established methods, PPO and PPO-Lagrangian; it does not propose a new constrained-RL algorithm. The contribution claimed is the comparison protocol and what it finds, not a new optimiser.
- **Hardware validation.** All results are produced in simulation. No policy from this thesis is run on the physical UR5e. That step belongs to Layer 3, out of scope here and named as future work in Chapter 6 rather than as a claim already tested.